

9.0 SUMMARY OF THE PROPOSED SYSTEM

The UTM Transport System is designed to resolve five campus transit pain points, they are bus arrival uncertainty, peak-hour seat shortages, unsafe carpooling, lack of route planning tools, and the absence of a proper feedback system. This project introduces a system to solve these problems. It includes secure user registration, live bus tracking with estimated time of arrival (ETA), real-time seat monitoring, official carpooling feature, smart route navigation and digital feedback system.

The system design ensures strict data boundaries and smooth information flow. User verification takes place through the internal registry. Real-time vehicle location updates continuously from GPS transponders. Passenger occupancy levels are tracked automatically via seat sensors. Moreover, the system provides matching rides for students and drivers. This system also integrated with google maps service to calculate the shortest route. All main storage units such as Carpool trip Database, Bus schedule Database and so on are designed consistently according to the overall architecture.

In conclusion, this system successfully solves the challenges posed by everyday delays in transit and administrative blind spots. By standardizing core processes, the system efficiently translates row information from users and vehicles into accurate tracking and monthly performance checks. Ultimately, it is hoped that this application will not only minimize transit anxiety but also foster a more efficient campus community for all UTM students.

